Separating endpoint and history information in radial and astrometric response design

HTT conditional response-design research

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Abstract

A radial velocity response measures the difference between source bulk motion and observer motion. A time-integrated channel also loses any history mode orthogonal to its window. We first construct a finite example in which independently calibrated observer and history responses remove these two ambiguities. Thirteen radial measurements at two radii determine twelve combinations of eighteen signed coefficients after removing a scalar calibration. Either additional three-component response raises the rank to fifteen; together they give rank eighteen. Equal radii reduce the joint rank to fourteen. We then examine a proper-motion response whose five-component trace-free tensor combines a source–observer endpoint difference with the present observer-time derivative. The associated ten-coordinate history map has rank five even when the observable tensor is fully determined. For a retained sample of 755,850 astrometric sources, we compare six common spin/glide nuisance coefficients with all 97 coefficients outside the common electric-quadrupole signal span, using the same complete normal matrix and score. Both projected tensor responses have rank five. Under the stipulated Gaussian noise law, their worst-direction tensor radii for 80 percent power are 2.2943 and 5.0499 microarcseconds per year, respectively. These are conditional design sensitivities. Their interpretation as matter shear requires a physical distance/history response, an optical-centroid transfer and a calibrated spatial error model.

I. A COMMON FRAME AND A FINITE RESPONSE

Consider a nonrotating orthonormal spatial frame carried by an expanding fiducial normal observer. We use signature $(-+++)$, proper radial positions rn, $n \cdot n = 1$, and the fixed scales

$$\begin{aligned} c &= 299792.458\ \mathrm{km\,s^{-1}}, & R_* &= 100\ \mathrm{Mpc}, \\ v_* &= 300\ \mathrm{km\,s^{-1}}, & \beta_* &= v_*/c, \\ h_* &= v_*/R_* = 3\ \mathrm{km\,s^{-1}\,Mpc^{-1}}. \end{aligned} \tag{1}$$

Our dimensionless radial datum is $y = (c/v_*)\delta z$. All vector components and tensors below refer to this same frame. The source bulk and observer velocities are $v_* b$ and $v_* \ell$. A positive observer velocity along the observer-to-source direction reduces the first-order radial redshift, giving the relative-motion term $n \cdot (b - \ell)$. A scalar rate coefficient $h_* m$

and a trace-free symmetric rate h_*S complete the local affine part of the response. These are coefficients of a local first-order design; the reference frame and scales do not supply a finite-redshift propagation law or an absolute background expansion measurement.

Choose the Frobenius-orthonormal STF basis

$$\begin{aligned} F_1 &= \frac{\mathrm{diag}(1,-1,0)}{\sqrt{2}}, \quad F_2 = \frac{\mathrm{diag}(1,1,-2)}{\sqrt{6}}, \\ F_3 &= \frac{e_x e_y^T + e_y e_x^T}{\sqrt{2}}, \qquad F_4 = \frac{e_x e_z^T + e_z e_x^T}{\sqrt{2}}, \qquad F_5 = \frac{e_y e_z^T + e_z e_y^T}{\sqrt{2}}. \end{aligned} \tag{2}$$

Thus $S = \sum_{a=1}^{5} s_a F_a$ and $\|S\|_F^2 = \sum_a s_a^2$. We retain all five signed coordinates and the full matrix, rather than its norm alone.

Let $\tau \in [0,1]$ parametrize a stipulated two-mode matter-tilt history,

$$\beta_G(\tau) = \beta_*[g_0 + (2\tau - 1)g_1]. \tag{3}$$

The coefficients g_0, g_1 are dimensionless vectors in the transported common frame. For a normalized window W, prescribe the dimensionless sensitivity

$$\rho n \cdot \int_0^1 W(\tau) \frac{\beta_G(\tau)}{\beta_*}\, d\tau, \qquad \rho = \frac{r}{R_*}. \tag{4}$$

Equation (4) defines this experiment. It does not derive a universal relation between tilt, acceleration and a redshift-rate dipole. A physical application must provide that relation and its window independently. The illustrative coefficient box $[-1,1]^{18}$ ensures $\sup_\tau \|\beta_G(\tau)\| \le 2\sqrt{3}\beta_* < 1$; the box serves only to locate the finite examples and is not a probability distribution or a cosmological admissible region.

Order the parameters as

$$\theta = (\ell_x, \ell_y, \ell_z, b_x, b_y, b_z, m, g_{0x}, g_{0y}, g_{0z}, s_1, \ldots, s_5, g_{1x}, g_{1y}, g_{1z})^T. \tag{5}$$

For the baseline window $W_0 = 1$, the integral of $2\tau - 1$ vanishes. With one free scalar calibration α, the thirteen baseline means are

$$y_i = \alpha + n_i \cdot (b - \ell) + \rho_i[m + n_i \cdot g_0 + n_i^T S n_i]. \tag{6}$$

The first nine directions, all at $\rho = 1$, are

$$e_x, -e_x, e_y, -e_y, e_z, -e_z, (3/5, 4/5, 0), (3/5, 0, 4/5), (0, 3/5, 4/5). \tag{7}$$

Append $e_x, -e_x, e_y, e_z$ at $\rho = q > 0$. The nominal design uses $q = 2$. Define E_3 to have rows n_i^T, define Q_5 to have rows $(n_i^T F_a n_i)_{a=1}^5$, and set $R = \mathrm{diag}(\rho_i)$. The baseline design and nuisance column are

$$J_0 = [-E_3,\ E_3,\ R\mathbf{1},\ RE_3,\ RQ_5,\ 0_{13\times 3}], \qquad N_0 = \mathbf{1}_{13}. \tag{8}$$

Here $R\mathbf{1}$ is a vector of dimensionless radii; it is distinct from the physical length R_*.

II. ONE NUISANCE QUOTIENT AND TWO ADDITIONAL RESPONSES

Two independently calibrated three-component responses are proposed:

$$z_L = \ell + e_L, \qquad z_G = \int_0^1 2\tau\, \frac{\beta_G(\tau)}{\beta_*}\, d\tau + e_G = g_0 + \frac{g_1}{3} + e_G. \tag{9}$$

The first supplies an observer-velocity anchor. The second uses the normalized window $W_1 = 2\tau$, whose overlap with $2\tau - 1$ is $1/3$. Their declared gains express both in dimensionless y-equivalent units. This is a pair of hypothetical measurement functionals, rather than a claimed delivered survey channel. Their calibration is independent in the algebraic sense that neither carries the free α of (6). No stochastic independence is assumed.

The augmented rows are

$$J_L = [I_3, 0, 0, 0, 0, 0], \qquad J_G = [0, 0, 0, I_3, 0, I_3/3], \qquad J = \begin{pmatrix} J_0 \\ J_L \\ J_G \end{pmatrix}, \qquad N = \begin{pmatrix} \mathbf{1}_{13} \\ 0_3 \\ 0_3 \end{pmatrix}. \tag{10}$$

Block widths in (10) follow (5). Removing an addition deletes its three rows from both J and N. For every design, the scalar nuisance is removed once, using the complete retained data vector and the same construction for all parameter columns. For a positive row-precision matrix W, define

$$P = I - \frac{(WN)(WN)^T}{\|WN\|^2}, \qquad A_{\mathrm{phys}} = PWJ, \qquad \bar{y} = PWy. \tag{11}$$

For the full design we take

$$W = \mathrm{diag}(I_{13}, \sigma_L^{-1} I_3, \sigma_G^{-1} I_3), \tag{12}$$

with nominal $\sigma_L = \sigma_G = 1$. The subdesigns retain the applicable diagonal blocks. These positive scales define a deterministic comparison metric. A physical joint sampling covariance and a statistical law are not supplied. In particular, the singular-value plots below do not carry confidence levels.

The kernel characterization

$$PWJ\theta = 0 \quad \Longleftrightarrow \quad J\theta \in \mathrm{col}\, N \tag{13}$$

holds because W is invertible and P projects away exactly WN. Thus finite positive precision weights alter conditioning but not structural identifiability. Allowing unknown offsets for the added responses would enlarge N and require a new analysis.

III. EXACT RANKS AND IDENTIFIED COMBINATIONS

Proposition 1 (two-radius separation). For (7), the nine-dimensional spherical polynomial space consisting of a monopole, a dipole and an STF quadrupole is unisolvent. With the four repeated directions in (7) placed at any distinct positive radius $q \neq 1$, the matrix

$$X = [E_4, RE_9], \qquad E_4 = [\mathbf{1}, E_3], \quad E_9 = [\mathbf{1}, E_3, Q_5] \tag{14}$$

has rank thirteen.

Proof. Let $f(n) = a + d \cdot n + n^T T n$, T STF, vanish on the first nine directions. The opposite axis pairs first give $d = 0$, then $a + T_{ii} = 0$. Taking the trace gives $a = 0$ and all three diagonal entries vanish. At the three remaining directions the nonzero cross terms force $T_{xy} = T_{xz} = T_{yz} = 0$. Hence E_9 is invertible. A degree-one polynomial vanishing at $e_x, -e_x, e_y, e_z$ is likewise zero, so the repeated-direction E_4 is invertible. A null vector of (14) represents polynomials $p_1 + \rho p_2$. At $\rho = 1$, unisolvence gives $p_2 = -p_1$ as polynomials. At the four repeated directions the equation becomes $(1 - q)p_1 = 0$, forcing $p_1 = p_2 = 0$. This is the retained two-radius rank argument, expressed in the orthonormal tensor coordinates (2). □

Proposition 2 (baseline quotient). At $q \neq 1$, the baseline response after (11) has rank twelve. Its exact six-dimensional kernel is

$$k_{\mathrm{split}}(v) = (v, v, 0, 0, 0, 0), \qquad k_{\mathrm{hist}}(w) = (0, 0, 0, 0, 0, w), \qquad v, w \in \mathbb{R}^3. \tag{15}$$

The identified signed combinations are

$$b - \ell, \qquad m, \qquad g_0, \qquad (s_1, \ldots, s_5). \tag{16}$$

Proof. By (13), a quotient-null vector satisfies $J_0\theta = a\mathbf{1}$ for some a. Proposition 1 applied to the coefficients $(-a, b-\ell, m, g_0, s)$ forces $a = 0$, $b - \ell = 0$, $m = g_0 = s = 0$. No row contains g_1. These are precisely (15), proving both completeness and the rank. □

The ambiguity in (15) has two separate mechanisms. A shared change to source bulk and observer velocity leaves their relative radial velocity unchanged. A history proportional to $2\tau - 1$ has zero unweighted mean. Repeated angular or radial observations with the same response window cannot reveal that history coefficient.

Proposition 3 (constructive augmentation). At $q \neq 1$, either response in (9) gives quotient rank fifteen and a three-dimensional kernel. Their joint addition gives quotient rank eighteen and a unique eighteen-coordinate inverse on the quotient data image.

Proof. The first addition acts on (15) as $J_L k_{\mathrm{split}}(v) = v$ and $J_L k_{\mathrm{hist}}(w) = 0$. The second acts as $J_G k_{\mathrm{split}}(v) = 0$ and $J_G k_{\mathrm{hist}}(w) = w/3$. Hence each addition removes exactly its corresponding three-dimensional kernel, and together they remove all six dimensions. Equation (13) extends the result to every positive finite precision weighting in (12). □

This count separates information supplied by an endpoint anchor from information supplied by a new history moment. The same conclusion follows by reconstructing

$$\ell = z_L, \qquad b = (b - \ell) + z_L, \qquad g_1 = 3(z_G - g_0), \tag{17}$$

after solving for (16).

Proposition 4 (equal-radius negative control). At $q = 1$, the baseline quotient has rank eight. Either addition has rank eleven, and their joint addition has rank fourteen.

Proof. At one radius, (6) depends only on the scalar $\alpha+m$, the dipole $b-\ell+g_0$, and the five STF coefficients. The nuisance quotient removes the scalar, leaving the independent dipole

and quadrupole, hence rank eight. Each calibrated addition contributes three independent output coordinates. With both additions the remaining kernel is generated by

$$\delta\theta_d = (0, -d, 0, d, 0, -3d), \qquad \delta\theta_m = (0, 0, t, 0, 0, 0), \quad d \in \mathbb{R}^3, t \in \mathbb{R}. \tag{18}$$

The first leaves every raw mean, including both additions, unchanged. The second is absorbed by $\delta\alpha = -t$. Conversely, a joint quotient-null vector obeys $\ell = 0$, $g_1 = -3g_0$, $b = -g_0$ and $s = 0$, with m free. Thus the kernel has exactly four dimensions. □

Equation (18) exhibits why independent endpoint and history anchors do not replace radial leverage. Distinct radii separate an angular intercept from a depth-proportional dipole and monopole.

IV. EXPLICIT FINITE WITNESSES

Starting from $\theta = 0$, choose $v = e_x/4$ and $w = e_y/4$. The two states $k_{\text{split}}(v)$ and $k_{\text{hist}}(w)$ both remain in the illustrative box. Every baseline prediction is exactly zero for each state. The first changes z_L by $(1/4, 0, 0)$; the second changes z_G by $(0, 1/12, 0)$. Their sum changes both additions while leaving all thirteen baseline means unchanged. These signed nineteen-coordinate predicted arrays are retained in the numerical supplement.

The history witness corresponds to

$$\beta_G(\tau) = \frac{\beta_*}{4}(2\tau - 1)e_y. \tag{19}$$

Its time integral vanishes although the history is nonzero except at its midpoint. The zero baseline column is therefore a response null direction, not a statement that the physical history vanishes.

For an equal-radius joint witness in the same box, take $d = e_x/4$ in (18): $b = -e_x/4$, $g_0 = e_x/4$, $g_1 = -3e_x/4$, with all other coordinates zero. All nineteen raw means remain zero. A separate change $m = 1/4$, $\alpha = -1/4$ supplies the monopole witness.

V. FIXED NORMALIZATION AND SPECTRAL COMPARISON

Velocity and rate scales in (1) are specified before numerical normalization. Let J_* and P_* denote the full design at $q = 2$, $\sigma_L = \sigma_G = 1$. Define once

$$d_j = \|P_* J_* e_j\|_2 > 0, \qquad L = \mathrm{diag}(d_1^{-1}, \ldots, d_{18}^{-1}), \qquad \theta = Lz. \tag{20}$$

The normalized response is

$$A = PWJL. \tag{21}$$

Nominal augmented columns have unit norm. The same numerical matrix L is then frozen for every subdesign, radius and precision case. In particular, the baseline's three zero history columns remain zero; weaker channels are not restored to unit norm during a scan. The supplement records the complete d_j, L, unscaled and scaled spectra and all physical unit conversions.

With $A = U\Sigma V^T$, the identified scaled coordinates are $V_r^T z = V_r^T L^{-1}\theta$, where V_r contains the right singular vectors for the nonzero spectrum. The corresponding projector is $\Pi_{\mathrm{id}} = V_r V_r^T$, and the kernel projector is $I - \Pi_{\mathrm{id}}$. The named physical combinations (16) span the same identified information but are not an orthonormal singular-vector basis. Singular-vector signs and bases within repeated singular subspaces have no physical significance.

The baseline is a 13×18 matrix of rank twelve. A thin SVD returns thirteen singular values and therefore only one explicit zero. The parameter kernel still has six dimensions. The complete eighteen-dimensional right basis and kernel projector are required to retain the five additional null directions absent from that thin list. The plotted spectrum may pad the missing five entries with structural zeros provided the caption identifies the padding.

The finite comparison uses the following analytic targets:

Design	Rows	Rank at $q = 2$	Kernel dimension	Rank at $q = 1$
Baseline	13	12	6	8
Baseline plus endpoint	16	15	3	11
Baseline plus history	16	15	3	11
Both additions	19	18	0	14

Design	Rank	Kernel	Positive smin	Identified condition	Full-vector gain
baseline	12	6	0.10665269	16.950054	unbounded

Design	Rank	Kernel	Positive smin	Identified condition	Full-vector gain
endpoint	15	3	0.097638737	18.655425	unbounded
history	15	3	0.10180602	17.900956	unbounded
both	18	0	0.093938751	19.539575	10.6452

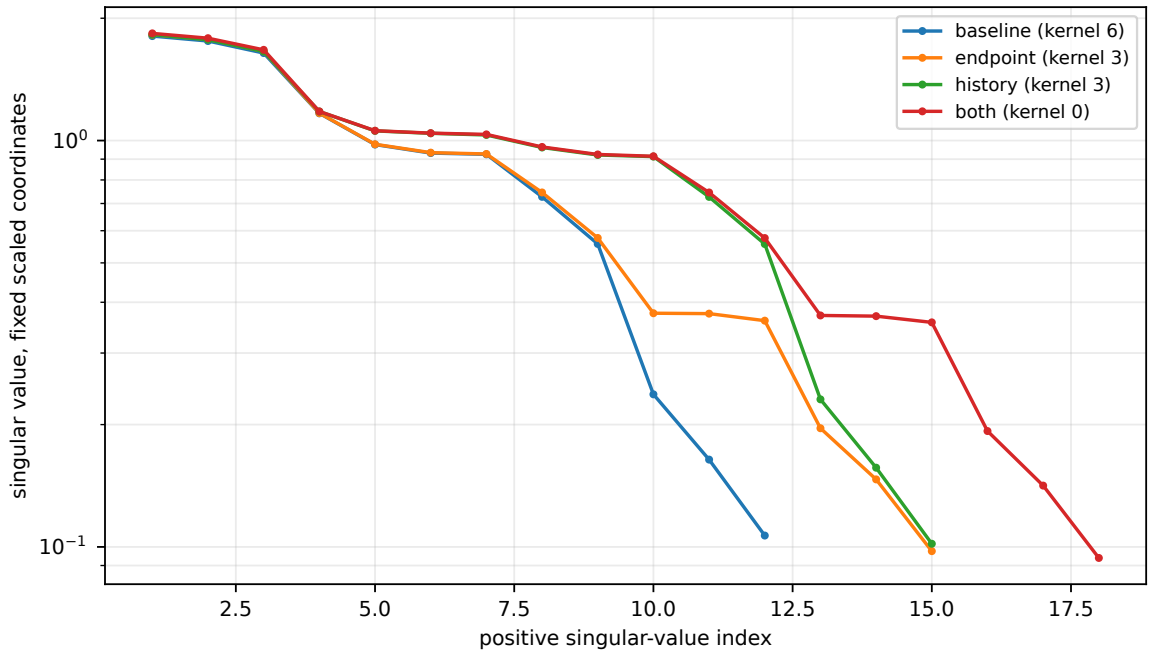


FIG. 1. Singular spectra at q=2 with the fixed normalization (20). Legends state structural null dimensions. The row metric is stipulated, not measured covariance.

To show radial sensitivity, set $q = 1 + \delta$ for

$$\delta \in \{0, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 3\}. \tag{22}$$

At every positive separation, the analytic ranks are those of Proposition 3; as the separation approaches zero, four joint singular directions become weak and at equality join the exact kernel. Numerical threshold dependence near equality should be displayed alongside the structural rank. A finite threshold does not change the algebraic theorem. At equality the inverse gain for all eighteen coordinates is infinite, even though the fourteen identified directions retain finite gains.

For additional-channel precision, evaluate every pair

$$(\sigma_L, \sigma_G) \in \{1/2, 1, 2, 4\}^2 \tag{23}$$

at $q = 2$, preserving (20). These cases measure how the weakest recoverable combination responds to the relative precision of the two added channels.

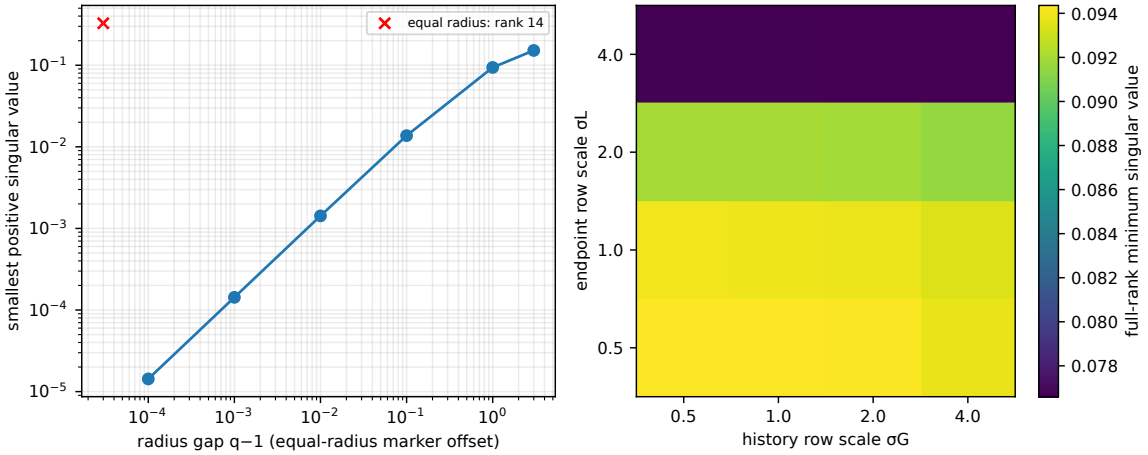


FIG. 2. Radial leverage and additional-channel precision in fixed scaled coordinates. The equal-radius marker is offset horizontally for log display; its full-vector inverse gain is unbounded.

VI. DETERMINISTIC ERROR SENSITIVITY

Let the normalized projected data obey $\bar{y} = Az + e$, with $\|e\|_2 \le \epsilon$. Suppose the operator used for inversion is $\widetilde{A} = A + \Delta$, where $\|\Delta\|_2 \le \eta$. Both errors are measured after the prescribed quotient and normalization. Thus Δ must include any error in that effective operator; nuisance-model changes cannot be treated as harmless coordinate changes. For a full-column-rank A, Weyl's bound gives

$$s_{\min}(\widetilde{A}) \ge s_{\min}(A) - \eta. \tag{24}$$

If $s_{\min}(A) > \eta$, the least-squares inverse satisfies

$$\|\widehat{z} - z\|_2 \le \frac{\epsilon + \eta\|z\|_2}{s_{\min}(A) - \eta}. \tag{25}$$

Proof. Full column rank of $\widetilde{A}$ gives $\widehat{z} - z = \widetilde{A}^{+}(e - \Delta z)$. Taking norms and using (24) proves (25). No distribution or independence assumption is used. □

The physical coefficient error obeys $\|\widehat{\theta} - \theta\|_2 \le \|L\|_2 \|\widehat{z} - z\|_2$, and any specified linear physical functional can use its own sharper factor $\|CL\|_2$. The illustration $\epsilon = 10^{-3}$, $\|z\|_2 \le 1$ is a separate error scenario, not the claim that the whole box $[-1, 1]^{18}$ lies in the unit z-ball. For $\eta/s_{\min} \in \{0, 0.1, 0.25, 0.5, 0.75, 0.9, 1, 1.1\}$, the bound loses certification at one. Such cases are reported as outside the certified domain.

For a fixed rank-deficient operator, the minimum-norm estimate recovers the orthogonal identified projection with

$$\|\widehat{z} - \Pi_{\mathrm{id}} z\|_2 \le \frac{\epsilon}{s^{+}_{\min}(A)}, \tag{26}$$

where $s^{+}_{\min}$ is the smallest nonzero singular value. This controls identified combinations only. An arbitrary kernel component leaves the data unchanged, so no uniform full-vector bound follows. Perturbations that change rank or the kernel require their own subspace control; (26) is the fixed-operator statement.

Equation (17) also gives direct, physically transparent bounds. If the errors on the estimated combinations obey $\|\delta(b - \ell)\| \le e_D$, $\|\delta z_L\| \le e_L$, $\|\delta g_0\| \le e_0$, and $\|\delta z_G\| \le e_G$, then

$$\|\delta b\| \le e_D + e_L, \qquad \|\delta g_1\| \le 3(e_G + e_0). \tag{27}$$

The history inverse incurs a factor of three because the new window overlaps the previously invisible temporal mode by only 1/3. This penalty survives the fact that the joint design is injective.

VII. A PROPER-MOTION ENDPOINT AND HISTORY RESPONSE

A. Signed response and its physical domain

The finite radial construction shows how an additional response can reveal a previously invisible history moment. A proper-motion observable provides a related example in which the endpoint and derivative contributions are already present in the same measured direction

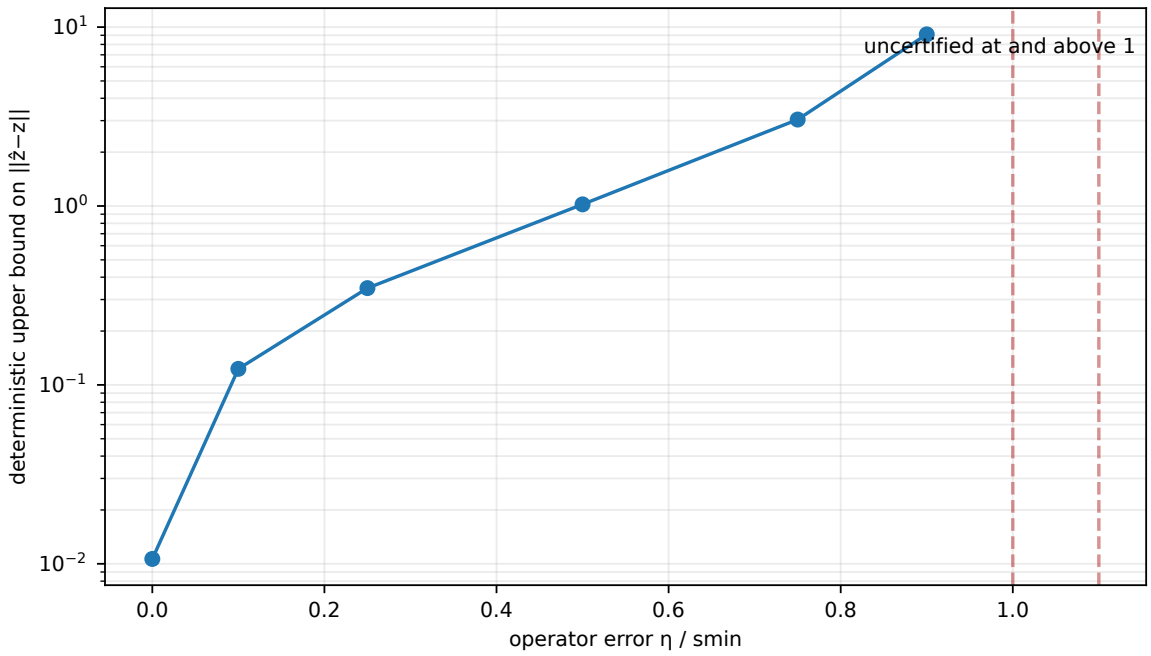


FIG. 3. Deterministic inverse bound (25) for epsilon=0.001 and ||z||<=1. At and above eta/smin=1 the bound is uncertified. No sampling law is supplied.

drift. To specify its domain, consider a weak homogeneous Bianchi I metric with fixed principal axes,

$$ds^2 = -c^2dt^2 + a(t)^2[e^{2B(t)}]_{ij}dx^i dx^j, \qquad B^T = B, \quad \operatorname{tr} B = 0. \tag{28}$$

Here B is dimensionless, t is cosmic proper time and the signature is $(-,+,+,+)$. Source and observer are comoving. We express the outward source direction n in a nonrotating orthonormal observer frame. To first order in the weak anisotropy, the direction-drift formula of Marcori et al. [1] [their Eq. (5.13)] becomes

$$\begin{aligned} D_{so} &= a_o c \int_{t_s}^{t_o} \frac{dt}{a(t)} > 0, \qquad P_n = I - nn^T, \\ \mu(n) &= P_n A_{\mathrm{PM}} n, \qquad A_{\mathrm{PM}} = \frac{2c}{D_{so}}(B_o - B_s) - \dot{B}_o. \end{aligned} \tag{29}$$

The two terms are the parallax contribution from the endpoint difference and the local aberration contribution. They have the same physical rate units. With c in kilometres per second and D in kilometres, A_{PM} has units of inverse seconds; the catalogue calculation converts radians per second to microarcseconds per Julian year of 365.25 days. The two

catalogue components are $(\mu_\alpha \cos\delta, \mu_\delta)$ in the ICRS basis. Identifying that basis with the transported model frame is a stated response convention, not a calibration of matter motion.

For a nearby affine separation, $B_o - B_s = \sigma_o \Delta t + O(\Delta t^2)$, $D_{so} = c\Delta t + O(\Delta t^2)$, and $\sigma_o = \dot{B}_o$. Equation (29) therefore gives $\mu = P_n \sigma_o n + O(\Delta t)$. The sign is positive. Differentiating the normalized separation when $\dot{r} = (HI + \sigma_o) r$ gives the same sign directly. In contrast, when the endpoint contribution is negligible at large distance, (29) approaches $-P_n \sigma_o n$. A large reported redshift alone does not establish that premise.

The accuracy of the nearby limit can be bounded without choosing a dynamical evolution. If $\|\dot{\sigma}(t)\|_F \le L_\sigma$ throughout the source–observer interval, the retained first-order response satisfies

$$\|A_{\rm PM} - \sigma_o\|_F \le 2 \left| \frac{c\Delta t}{D_{so}} - 1 \right| \|\sigma_o\|_F + \frac{c L_\sigma \Delta t^2}{D_{so}}. \tag{30}$$

This is a bound within the linear response, rather than a bound on omitted higher-order metric terms. It requires D_{so}, Δt and L_σ. These inputs are not supplied by the common-field catalogue statistics.

Equal endpoint deformations also need not give equal direction drifts. For a fixed STF tensor T, the kinematic perturbation $\delta B = \varepsilon \tau (1-\tau) T$, with $\tau = (t - t_s)/\Delta t$, leaves both endpoints unchanged and changes the present derivative by $-\varepsilon T/\Delta t$. Hence it changes $A_{\rm PM}$ by $+\varepsilon T/\Delta t$. This witness imposes no field equation or matter closure.

B. Observable tensor and unresolved histories

Expand $A_{\rm PM} = \sum_{a=1}^{5} a_a F_a$ in (2), retaining the signed coefficient vector $\boldsymbol{a}$. A common $A_{\rm PM}$ produces an electric quadrupole. Its full-sphere mean square is

$$\frac{1}{4\pi} \int |P_n A_{\rm PM} n|^2 \, d\Omega = \frac{\|A_{\rm PM}\|_F^2}{5} = \frac{\boldsymbol{a}^T \boldsymbol{a}}{5}. \tag{31}$$

The integral is an analytic sphere average. It is not the average over the selected catalogue. The norm alone also loses the signed tensor pattern.

For a declared common distance, choose a positive reference time $T_{\rm ref}$ and express $(B_o - B_s)/T_{\rm ref}$ and $\dot{B}_o$ in the same rate units, with five-vectors $\boldsymbol{q}_H$ and $\boldsymbol{s}_H$. Then

$$\boldsymbol{a} = R_\kappa \begin{pmatrix} \boldsymbol{q}_H \\ \boldsymbol{s}_H \end{pmatrix}, \qquad R_\kappa = [\kappa I_5, -I_5], \qquad \kappa = \frac{2cT_{\mathrm{ref}}}{D_{so}}. \tag{32}$$

The five exact null directions obey $\delta \boldsymbol{s}_H = \kappa \delta \boldsymbol{q}_H$. If $\mathcal{I}$ is the information for the observable tensor, the complete history information is

$$\mathcal{I}_H = R_\kappa^T \mathcal{I} R_\kappa = \begin{pmatrix} \kappa^2 \mathcal{I} & -\kappa \mathcal{I} \\ -\kappa \mathcal{I} & \mathcal{I} \end{pmatrix}. \tag{33}$$

Its rank remains five when $\mathcal{I}$ has rank five. The off-diagonal blocks cannot be omitted. Resolving B_o and B_s separately introduces the additional common-endpoint invariance $\delta B_o = \delta B_s$. A time-independent deformation can represent a spatial-coordinate rescaling within the weak fixed-axis metric; this coordinate freedom is distinct from the unresolved endpoint–derivative histories in (32). The ten-coordinate map describes an enlarged kinematic parameter space. A fixed-axis physical history may impose further restrictions, including common principal axes, and requires a separate model. The catalogue fit does not select one.

As an explicit conditional family, set $\boldsymbol{q}_H = \boldsymbol{s}_H$. The signed gains at $\kappa = 0, 0.1, 1, 2, 10$ are $-1, -0.9, 0, 1, 9$. Their squared values multiply a quadratic signal strength. The cancellation at $\kappa = 1$ belongs to this chosen family; it is not a universal absence of astrometric sensitivity. Likewise, the common-distance calculation does not assign a physical distance to every catalogue source.

C. A common-data nuisance comparison

We use the retained normal matrix and score for 755,850 selected astrometric sources. The 102-column design contains a common electric quadrupole and the other retained angular and template responses. Its working covariance includes each source's reported two-component marginal covariance and stipulates independence between different sources. It does not supply an empirical spatial covariance or a likelihood for the source selection.

Let X denote the retained design, C_w this working covariance, and y the measured proper-motion vector. The stored quantities are $G = X^T C_w^{-1} X$ and $\boldsymbol{b} = X^T C_w^{-1} y$. The five-component embedding J selects the complete common electric-quadrupole span and changes its coordinates to the Frobenius basis (2). For a nuisance embedding N, profiling gives

$$\begin{aligned}
\mathcal{I}_N &= J^T G J - J^T G N (N^T G N)^+ N^T G J, \\
\boldsymbol{h}_N &= J^T \boldsymbol{b} - J^T G N (N^T G N)^+ N^T \boldsymbol{b}, \\
\widehat{\boldsymbol{a}}_N &= \mathcal{I}_N^{-1} \boldsymbol{h}_N, \qquad C_{a,N} = \mathcal{I}_N^{-1}.
\end{aligned} \tag{34}$$

The inverse in the last line applies to the rank-five cases below. The numerical evaluation uses an orthogonal projection in whitened coordinates. It preserves the signal–nuisance cross terms, the complete five-dimensional covariance and the joint signal–nuisance covariance.

The first profile fits six common spin/glide coefficients. They are formed by summing the corresponding columns in the three disjoint reported-redshift bins. The second profiles all 97 original non-E2 columns. Both profiles use the same sample, coefficients, units, G and b; the second is a sensitivity calculation on the same realization. Enlarging the nuisance space reduces information in positive-semidefinite order. Full-sphere harmonic orthogonality cannot be used to discard cross terms on this selected, weighted sky.

Nuisance dimension	a_1	a_2	a_3	a_4	a_5
6	-2.262422	2.271772	-3.653029	-2.718689	2.395232
97	-5.727189	1.156649	-7.427496	-5.561747	6.176964

Signed common electric-quadrupole coefficients in the basis (2), in microarcseconds per year, from the same selected sample under two nuisance profiles. The complete covariances and signal–nuisance cross covariances accompany the numerical supplement. These coefficients describe the selected optical proper-motion field under the stipulated error law; they are not calibrated matter-shear estimates.

Nuisance	Rank	Information eigenvalues	Best / worst radius
6	5	2.43693, 2.65563, 3.17145, 3.67145, 4.95102	1.60963 / 2.29431
97	5	0.503015, 0.629927, 1.13946, 1.46083, 2.38294	2.32015 / 5.04990

Information eigenvalues have units $(\mu\mathrm{as\,yr}^{-1})^{-2}$; radii are in microarcseconds per year. Eigenvectors need not coincide between profiles.

The first profile has information eigenvalues $(2.43693, 2.65563, 3.17145, 3.67145, 4.95102)$ in $(\mu\mathrm{as\,yr}^{-1})^{-2}$. With all 97 nuisance coefficients they become $(0.503015, 0.629927, 1.13946, 1.46083, 2.38294)$. The common tensor remains identifiable in both working designs, although its least constrained direction becomes substantially weaker.

There is a useful limiting check. If an arbitrary common E2 nuisance is admitted, it spans exactly the signal image and the profiled information vanishes. A free common spin similarly absorbs a common magnetic dipole. The corresponding coefficient set is unbounded unless another domain restriction is supplied. It is not empty. This is a reference-field ambiguity, and a calibrated frame condition would need its own response and uncertainty.

D. Conditional detection power

For a stipulated Gaussian fixed-design experiment, a five-dimensional quadratic test has a central chi-square law under the zero-field null and a noncentral law under a specified coefficient vector. At size alpha=0.05, define

$$\lambda = \boldsymbol{a}^T \mathcal{I}_N \boldsymbol{a}, \qquad \mathcal{P}(\boldsymbol{a}) = \Pr\{\chi^2_5(\lambda) > \chi^2_{5,0.95}\}. \tag{35}$$

For a prescribed Frobenius radius $\rho = \|\boldsymbol{a}\|_2$, the largest and smallest noncentralities are $\rho^2\lambda_{\max}(\mathcal{I}_N)$ and $\rho^2\lambda_{\min}(\mathcal{I}_N)$. The retained calculation gives $\chi^2_{5,0.95} = 11.07049769$ and the noncentrality 12.82760657 for 80 percent power. The resulting best/worst radii are 1.60963/2.29431 microarcseconds per year for the six-coefficient profile and 2.32015/5.04990 for the 97-coefficient profile. At $\rho = 0$ the power is 0.05.

The Gaussian power calculation uses the actual selected angular coverage and marginal error weights. It does not justify a matter-shear limit from the selected coefficients. That step requires a source-dependent distance/history law, an optical-centroid-to-matter response, reference-frame calibration, selection modelling and a measured or justified spatial covariance. In particular, common-E2 sufficient statistics cannot generate arbitrary new depth-weighted E2 cross moments.

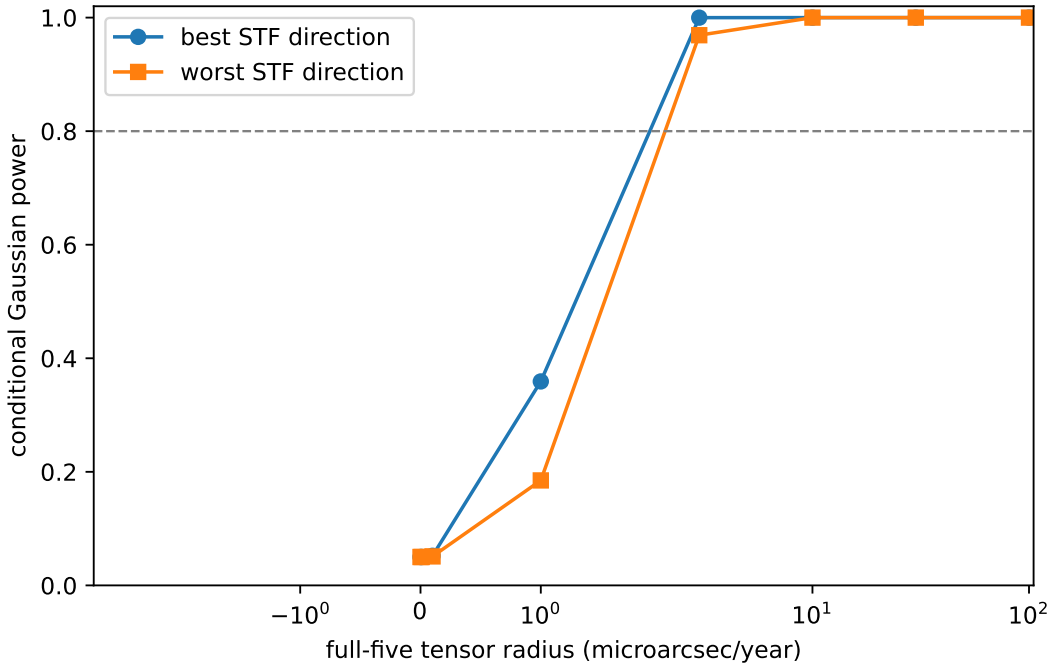


FIG. 4. Conditional Gaussian power for the common five-component proper-motion tensor after profiling six common spin/glide coefficients. The radius is the Frobenius norm in microarcseconds per year. Markers are retained evaluated radius points for the best and worst information eigenvectors; connecting segments are visual guides, not evaluated interpolation. The 80-percent radii come from the numerical table and Eq. (35), not crossings of those segments; the analytic full-sphere RMS is $\rho/\sqrt{5}$. The plotted probabilities use the reported marginal source errors and stipulated cross-source independence. They are design sensitivities, not empirical exclusion limits or a p-value for the observed field.

VIII. PHYSICAL INTERPRETATION AND THE NEXT MEASUREMENT

The radial and astrometric examples separate observable information from the physical history to which it may be attributed. In the finite radial design, an independently calibrated observer response removes the observer–bulk ambiguity, while a second temporal window removes a history moment invisible to the first. Radial leverage remains necessary: at equal radii four parameter combinations survive even after both additions. These conclusions follow from the stated response functions and common nuisance quotient.

For the astrometric example, the observable is a signed STF tensor A_{PM}. It combines an endpoint difference and a present derivative. The selected catalogue supports a rank-

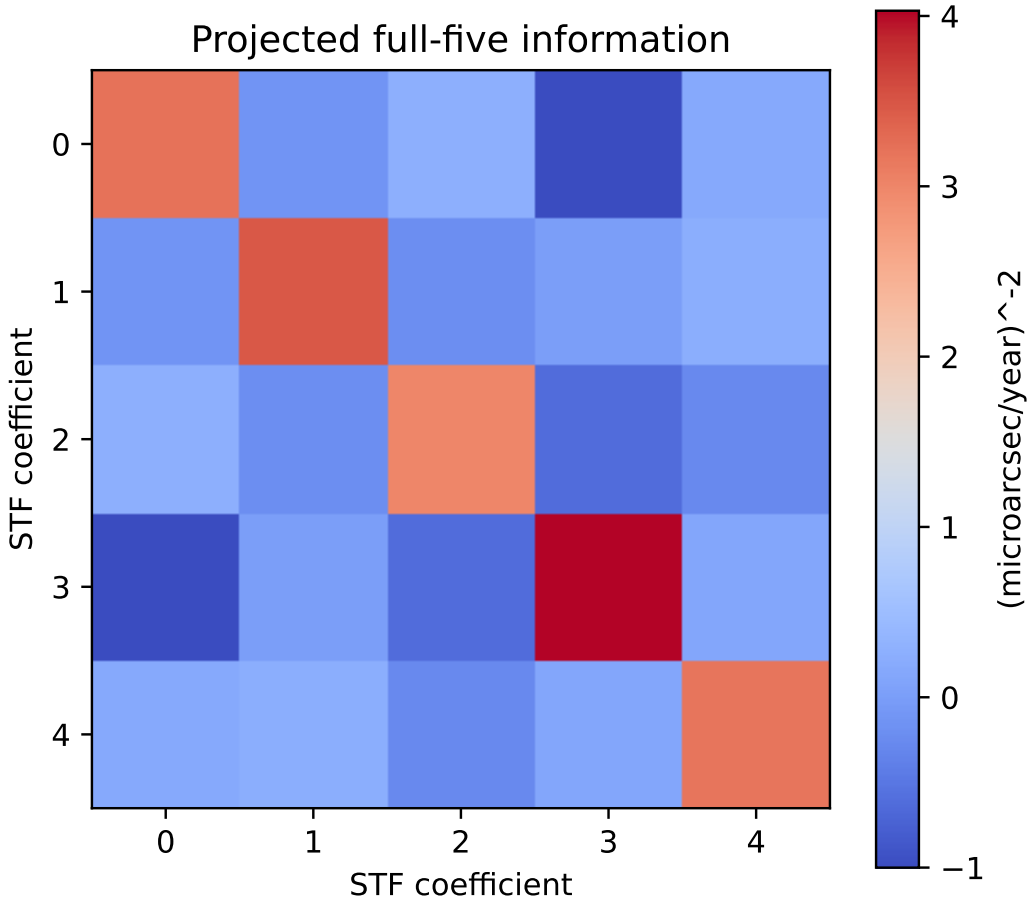


FIG. 5. Full five-component projected information for the six-coefficient common spin/glide profile, in the Frobenius basis (2), with units $(\mu\mathrm{as\,yr}^{-1})^{-2}$. Off-diagonal entries retain the coupling induced by selected sky coverage and working weights. The 97-coefficient sensitivity profile uses the same data and is compared in the accompanying table.

five common-field fit under both nuisance profiles, but the associated ten-coordinate history information retains five null directions. More precise measurements of the same common field do not remove them. A physically calibrated distance dependence or another independent response is needed. The numerical power frontier quantifies sensitivity to this observable tensor under its working noise law.

The complete carrier therefore retains the signed radial vectors, scalar rate, full STF tensors, response functions, joint covariance or error set, and the relevant sky and depth supports. The radial contraction cannot measure an antisymmetric velocity gradient be-

cause $n^T \Omega n = 0$. Matter vorticity remains unmeasured. A chosen hypersurface normal's irrotationality does not determine matter vorticity, and neither a fitted radial tensor nor an optical proper-motion quadrupole by itself supplies a calibrated matter-frame expansion/shear tensor.

A physical comparator x, including any declared signed projection x_C, requires its complete tensor components and frame. The normalized score Q additionally needs a physically defined anchor and denominator domain. The occupancy F needs a support measure and an admissible ceiling; the exceedance Pi needs its specified joint law; and the depth functional G_F needs compatible depth supports and their joint dependence. The results here do not supply those missing sectors. They remain unknown and are not assigned zero. Deterministic tensor images may nevertheless be reported within the stated response model.

The additional physical inputs are concrete: endpoint and history response kernels, source and observer frame transport, units, calibration and nuisance coupling, and a justified response error or joint covariance. The first-order Bianchi I expression illustrates one possible response under its own assumptions. Its use here does not identify a Bianchi family from the observed morphology.

Appendix A: Numerical appendix: reproducible comparison

The companion consumer verifies the sealed retained inputs, constructs (2), (7)–(12), freezes (20), and emits all designs, full right singular bases, kernel/identified projectors, spectra and signed witnesses. The nominal rank counts must agree at relative cutoffs $10^{-12}, 10^{-10}, 10^{-8}$. Frobenius-basis and null residuals must be at most 10^{-12}; reconstruction and deterministic bound checks use relative tolerance 10^{-10} or absolute tolerance 10^{-12} for unit-scale fixtures. Exact analytic rank remains authoritative near equal radii. The generated tables and figures report both numerical values and their design assumptions.

The supplementary source map records the retained rank argument, response conventions and numerical inputs. The astrometric supplement retains the full five-component coefficients, covariance, signal–nuisance cross covariance, information eigenbasis, ten-coordinate history information and exact response-null arrays. The two nuisance profiles consume the same 102-dimensional normal matrix and score; no old prior is imported. All displayed astrometric values and figures are generated from these retained quantities.

Appendix B: Generated numerical appendix tables

Parameter	Physical coefficient unit	d_j	L_jj
ell_x	v*=300 km/s	2.368381466	0.4222292795
ell_y	v*=300 km/s	2.134695078	0.4684509794
ell_z	v*=300 km/s	2.181742423	0.4583492485
b_x	v*=300 km/s	2.146911915	0.4657852952
b_y	v*=300 km/s	1.885980667	0.5302281287
b_z	v*=300 km/s	1.939071943	0.5157106231
m	h*=3 km/s/Mpc	1.664100589	0.6009252126
g0_x	beta*=300/299792.458	3.407232127	0.2934933584
g0_y	beta*=300/299792.458	2.666602563	0.3750090147
g0_z	beta*=300/299792.458	2.698717644	0.3705463601
s_1	h*=3 km/s/Mpc	2.8381359	0.3523439452
s_2	h*=3 km/s/Mpc	2.664247621	0.3753404872
s_3	h*=3 km/s/Mpc	0.6521915717	1.533291817
s_4	h*=3 km/s/Mpc	0.6521915717	1.533291817
s_5	h*=3 km/s/Mpc	0.6521915717	1.533291817
g1_x	beta*=300/299792.458	0.3333333333	3
g1_y	beta*=300/299792.458	0.3333333333	3
g1_z	beta*=300/299792.458	0.3333333333	3

1. Complete nominal scaled spectrum: baseline

Index	Singular value
1	1.80776894878
2	1.75783553205
3	1.64202991241
4	1.16693241497
5	0.977608148968

Index	Singular value
6	0.93153934285
7	0.925000426039
8	0.72695402196
9	0.556681575992
10	0.237422793074
11	0.164010989736
12	0.106652693887
13	2.28393497472e-16
14	structural zero (right completion)
15	structural zero (right completion)
16	structural zero (right completion)
17	structural zero (right completion)
18	structural zero (right completion)

2. Complete nominal scaled spectrum: endpoint

Index	Singular value
1	1.8214921072
2	1.77388823641
3	1.6579048791
4	1.16814162944
5	0.979221208186
6	0.933867897306
7	0.92679745181
8	0.745208194267
9	0.57589372613
10	0.375920041292
11	0.375052692147
12	0.360409875835

Index	Singular value
13	0.195900842657
14	0.146618339872
15	0.097638736537
16	0
17	structural zero (right completion)
18	structural zero (right completion)

3. Complete nominal scaled spectrum: history

Index	Singular value
1	1.8224251976
2	1.77075892464
3	1.6569969902
4	1.17889465675
5	1.05629533648
6	1.04195741167
7	1.03219924361
8	0.961460443345
9	0.921463617644
10	0.91312580769
11	0.726700364284
12	0.55634543484
13	0.230754137695
14	0.1566639275
15	0.101806023814
16	1.84960232996e-16
17	structural zero (right completion)
18	structural zero (right completion)

Index	Singular value

4. Complete nominal scaled spectrum: both

Index	Singular value
1	1.83552330269
2	1.78615342167
3	1.6720014412
4	1.17933809547
5	1.05743023017
6	1.04292085192
7	1.03553442499
8	0.964471288015
9	0.924256501433
10	0.914865452434
11	0.745162919861
12	0.57555461857
13	0.371204892219
14	0.36975411719
15	0.356918568169
16	0.192824411934
17	0.141550104788
18	0.0939387506842

All arrays, complete right bases and source-linked checks are in the companion JSON artifacts. This numerical appendix is generated from the frozen design and does not add

observed channels.

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[1] O. H. Marcori, C. Pitrou, J.-P. Uzan, and T. S. Pereira, "Direction and redshift drifts for general observers and their applications in cosmology," Phys. Rev. D **98**, 023517 (2018), arXiv:1805.12121.